1-3

Perform the following computations (i) exactly, (ii) using three-digit chopping arithmetic, and (iii) using three-digit rounding arithmetic. (iv) Compute the relative errors in parts (ii) and (iii).

a. $\frac{4}{5}+\frac{1}{3}$ b. $\left(\frac{1}{3}+\frac{3}{11}\right)-\frac{3}{20}$

a. 4/5 + 1/3

(i)exactly:

4/5 + 1/3 = 12/15 + 5/15 = 17/15

(ii)three-digit chopping arithmetic:

4/5 = 0.800

ch(1/3) = 0.333

0.800 + 0.333 = 1.133

ch(1.133) = 1.13

(iii)three-digit rounding arithmetic:

4/5 = 0.800

rd(1/3) = 0.333

0.800 + 0.333 = 1.133

rd(1.133) = 1.13

(iv)relative errors:

chppong:

| (17/15) - 1.13 | / | 17/15 | = | 1.1333... - 1.13 | / 1.1333... ≈ 0.00333 / 1.13333 ≈ 0.00294

rounding:

| (17/15) - 1.13 | / | 17/15 | = | 1.1333... - 1.13 | / 1.1333... ≈ 0.00333 / 1.13333 ≈ 0.00294

b. (1/3 + 3/11) - 3/20

(i)exactly:

(1/3 + 3/11) - 3/20 = (11/33 + 9/33) - 3/20 = 20/33 - 3/20 = (400 - 99)/660 = 301/660

(ii)three-digit chopping arithmetic:

ch(1/3) = 0.333

ch(3/11) = 0.272

0.333 + 0.272 = 0.605

3/20 = 0.150

0.605 - 0.150 = 0.455

(iii)three-digit rounding arithmetic:

rd(1/3) = 0.333

rd(3/11) = 0.273

0.333 + 0.273 = 0.606

3/20 = 0.150

0.606 - 0.150 = 0.456

(iv)relative errors:

chopping:

| (301/660) - 0.455 | / | 301/660 | = | 0.456060... - 0.455 | / 0.456060... ≈ 0.0010606 / 0.456060... ≈ 0.002326

rounding:

| (301/660) - 0.456 | / | 301/660 | = | 0.456060... - 0.456 | / 0.456060... ≈ 0.0000606 / 0.456060... ≈ 0.000133

1-5

Implement the Bisection method in C or matlab and find solutions accurate to within 10^{-5} for the following problems. **(List the midpoints in each iteration as well) .**

a. $e^x - x^2 + 3x - 2 = 0$ for $0 \le x \le 1$

b. $x\cos x - 2x^2 + 3x - 1 = 0$ for $0.2 \le x \le 0.3$ and $1.2 \le x \le 1.3$

代码输出：

--- Bisection for Problem a on [0.0, 1.0] ---

Iteration Midpoint (p)

---

1 0.5000000000
2 0.2500000000
3 0.3750000000
4 0.3125000000
5 0.2812500000
6 0.2656250000
7 0.2578125000
8 0.2617187500
9 0.2597656250
10 0.2587890625
11 0.2592773438
12 0.2590332031
13 0.2591552734
14 0.2590942383
15 0.2591247559
16 0.2591400146
17 0.2591323853

Approximate solution is p = 0.2591323853

--- Bisection for Problem b on [0.2, 0.3] ---

Iteration Midpoint (p)

---

1 0.2500000000
2 0.2750000000
3 0.2875000000
4 0.2937500000
5 0.2968750000
6 0.2984375000
7 0.2976562500

8 0.2972656250
9 0.2974609375
10 0.2975585938
11 0.2976074219
12 0.2975830078
13 0.2975708008
14 0.2975646973

Approximate solution is p = 0.2975646973

--- Bisection for Problem b on [1.2, 1.3] ---
Iteration Midpoint (p)
---
1 1.2500000000
2 1.2250000000
3 1.2375000000
4 1.2437500000
5 1.2406250000
6 1.2421875000
7 1.2429687500
8 1.2425781250
9 1.2427734375
10 1.2426757812
11 1.2427246094
12 1.2426999023
13 1.2427122559
14 1.2427060791

Approximate solution is p = 1.2427060791

1-6

Implement the fixed-point iteration method in C or matlab and find solutions accurate to within 10^{-2} for the following problems. **(List pn in each iteration as well)**.

a. $2\sin \pi x + x = 0$ on [1, 2], use $p_0 = 1$

b. $3x^2 - e^x = 0$

代码输出：
--- Fixed-Point Iteration for Problem a ---
Initial guess p0 = 1.00
Iteration p_n
---
1 1.16666667

2 1.25327090
3 1.29828239
4 1.32186769
5 1.33405364
6 1.34032333

Solution found: p = 1.34032333

--- Fixed-Point Iteration for Problem b ---
Initial guess p0 = 1.00
Iteration p_n
---
1 0.95189255
2 0.93240092
3 0.92317789

Solution found: p = 0.92317789

1-7

Let $g \in C^1[a,b]$ and p be in (a, b) with g(p) = p and $|g'(p)|>1$. Show that there exists a $\delta>0$ such that if $0<|p_0-p|<\delta$, then $|p_0-p|<|p_1-p|$. Thus, no matter how close the initial approximation p_0 is to p, the next iterate p_1 is father away, so the fixed-point iteration does not converge if $p_0 \neq p$.

考虑 p_1 与 p 的误差：
$p_1 - p = g(p_0) - g(p)$
由于 $g \in C^1[a,b]$，根据中值定理 (Mean Value Theorem)，$\exists \xi$ 介于 p_0 和 p 之间，使得：
$g(p_0) - g(p) = g'(\xi)(p_0 - p)$
因此，
$p_1 - p = g'(\xi)(p_0 - p)$
两边取绝对值，得到误差传播方程：
$|p_1 - p| = |g'(\xi)| |p_0 - p|$ (1)
令 $L = |g'(p)|$。已知 $L > 1$。
由于 g'在 p 点连续，根据极限的 ε-δ 定义，对于 $\varepsilon = (L - 1)/2 > 0$，$\exists \delta > 0$ 使得：
$|x - p| < \delta \Rightarrow |g'(x) - g'(p)| < \varepsilon$
对于满足 $|x - p| < \delta$ 的任意 x，应用反向三角不等式：
$|g'(x)| = |g'(p) - (g'(p) - g'(x))| \geq |g'(p)| - |g'(p) - g'(x)|$
$|g'(x)| > L - \varepsilon = L - (L - 1)/2 = (L + 1)/2$
因为 $L > 1$，所以 $(L + 1)/2 > 1$。
故，$|x - p| < \delta \Rightarrow |g'(x)| > 1$。
选择 p_0 使得 $0 < |p_0 - p| < \delta$。

由于 ξ 介于 p_0 和 p 之间，必然有 $|\xi - p| < |p_0 - p| < \delta$。
因此，根据第 5 步的结论，$|g'(\xi)| > 1$。
将 $|g'(\xi)| > 1$ 代入方程 (1)：
$|p_1 - p| = |g'(\xi)| |p_0 - p| > 1 \cdot |p_0 - p|$
最终得到：
$|p_1 - p| > |p_0 - p|$
结论：
若 $p_0 \neq p$ 且在 p 的 δ-邻域内，则 p_1 必比 p_0 更远离 p。因此，不动点迭代发散。